

B.Tech Project (CP303)



Title: Approval-Based Committee Selection with Fairness Constraints

Group: 15

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Multiwinner Voting

Multiwinner voting is the process of selecting a group of winners (a committee) instead of a single candidate.

Some classical multiwinner voting rules:

- AV(Approval Voting): Pick k candidates with the most approvals.
- PAV(Proportional Approval Voting): Ensures proportional representation.
- CAV(Chamberlin–Courant): Each voter gets a representative.
- SAV(Satisfaction Approval Voting): Maximize total satisfaction.
- NAV/NSAV: Approvals minus disapprovals

Recap

TEMPORAL COMMITTEE SELECTION FRAMEWORK:

We studied committee selection in a temporal setting where:

- committees are selected over multiple rounds/time steps
- voter approval preferences may change over time
- the goal is to balance welfare and fairness simultaneously

MAIN COMPONENTS OF THE FRAMEWORK:

1. Welfare Functions:

- Utilitarian welfare → maximize total voter satisfaction
- Egalitarian welfare → improve representation of the least satisfied voter

2. Welfare–Fairness Tradeoff:

We introduced fairness guarantees to ensure that:

- no voter remains persistently under-represented
- committee quality is maintained while improving fairness



ILP-BASED OPTIMIZATION:

We formulated the problem using Integer Linear Programming (ILP) to:

- optimize committee selection
- enforce fairness constraints
- maximize overall welfare



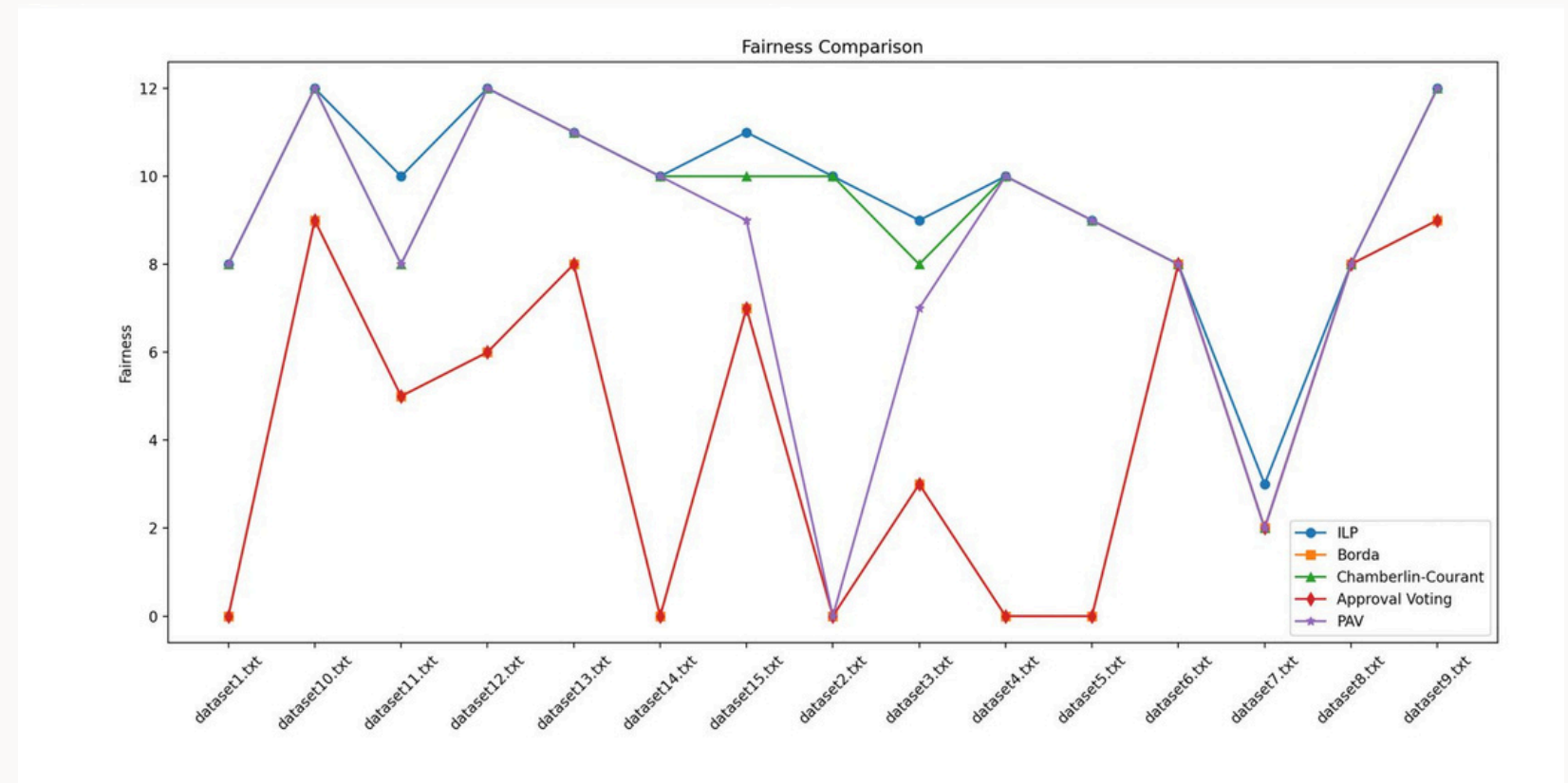
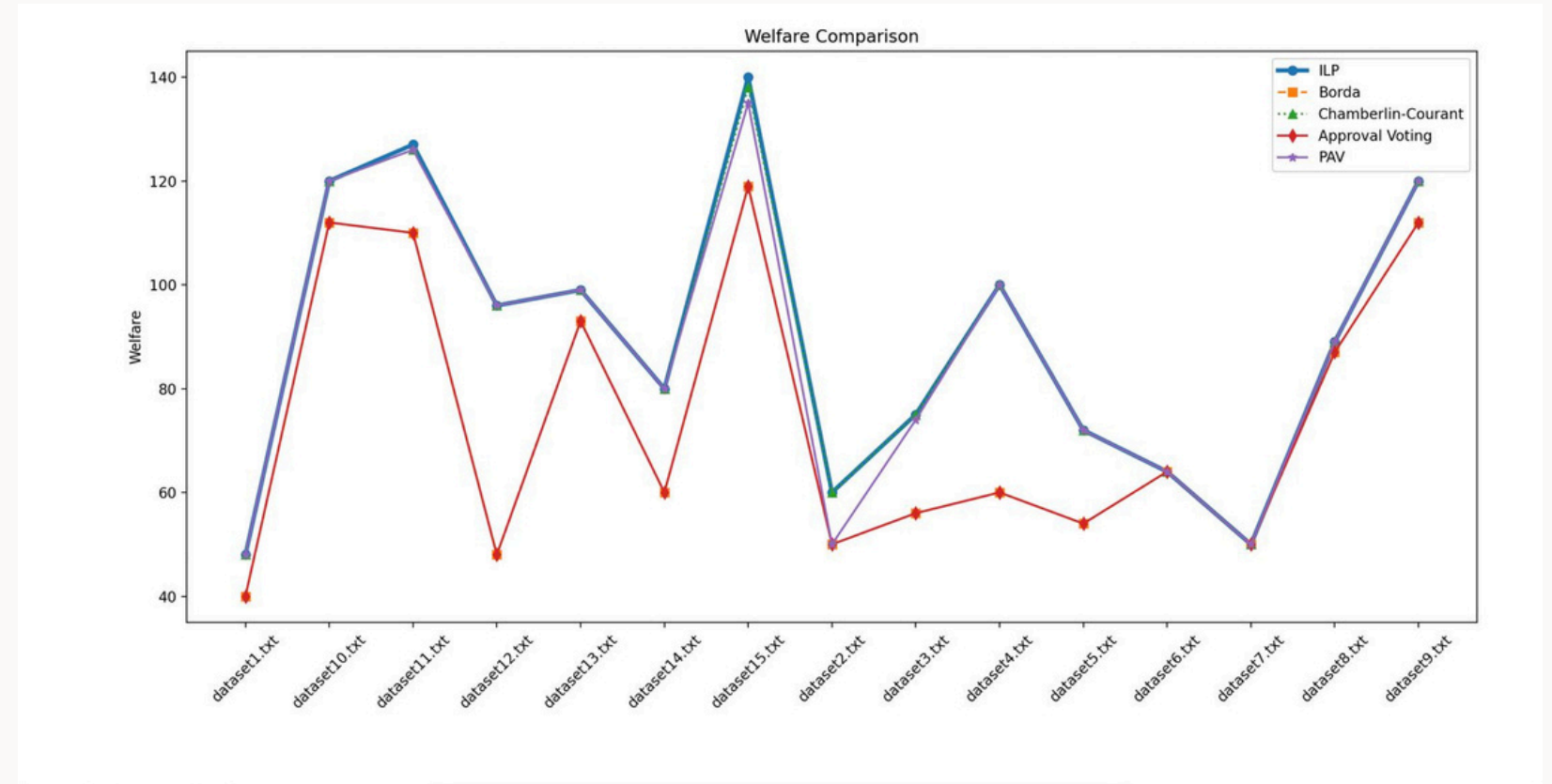
EXPERIMENTAL ANALYSIS

We compared our framework with:

- Approval Voting (AV)
- PAV
- Borda
- Chamberlin-Courant (CC)

using:

- welfare metrics
- fairness metrics



NEW SETTING:



LIMITATION OF THE EARLIER ILP FRAMEWORK:

Although the formulation produced good committee solutions, the approach had important limitations:

- **ILP solutions** become **computationally expensive** for larger instances.
- Exact optimization does not easily scale with increasing committee and voter sizes.
- The framework did not explicitly model realistic candidate attributes and quota-based constraints.



MOTIVATION FOR NEW SETTING:

Practical committee selection problems involve:

- Candidates with multiple attributes and quota requirements.
- Approval-based committee selection under explicit multi-level attribute constraints.
- Tradeoff between utility maximization $U(S)$ and fairness-aware committee construction
- Studying utility loss while satisfying attribute constraints
- Deterministic, partition-based and approximation approaches for NP-hard settings

Problem Settings

- We are given a set of candidates $C=\{c1,c2,...,cm\}$ and a set of voters $V=\{v1,v2,...,vn\}$.
- Each voter submits an approval ballot $A_i \subseteq C$ containing the candidates they approve of.
- The objective is to select a committee $W \subseteq C$, $|W| = k$ of fixed size k .
- Each candidate is associated with multiple protected attributes such as gender, race, region, or category, and fairness constraints are imposed on these protected groups.
- For every protected group G_j , $|W \cap G_j| \geq \ell_j$ where ℓ_j denotes the minimum required representation.

Problem Settings

The attributes are intersectional, meaning a single candidate may satisfy multiple protected attributes simultaneously (e.g., a White Female candidate contributes to both White and Female representation constraints).

The goal is to select a committee that:

- satisfies all fairness constraints, and
- maximizes the total approval utility of voters.

$$\max \sum_{i=1}^n |W \cap A_i|$$

NP-Hardness of the Problem

Approval Utility Maximization is NP-Hard

- Selecting a committee that maximizes total voter approval is computationally difficult.
- The algorithm may need to search among mC_k possible committees.

Fair Committee Selection is NP-Hard

- Selecting a committee satisfying multiple fairness constraints is also computationally difficult.
- Overlapping and intersectional protected attributes make the constraints interdependent.

Our Problem

- Our work combines approval utility maximization with fair committee selection.
- The objective is to maximize voter satisfaction while satisfying all fairness constraints.

Our Approach

1. Proportional Approval Fair Committee Selection (PAF-Comm):

PAF-Comm is an approval-based committee selection framework that jointly optimizes voter satisfaction and group fairness under protected and intersectional attributes.

CORE COMPONENTS:

- **Approval Utility**

We use Proportional Approval Voting (PAV) utility:

$$U(S) = \sum_{i=1}^n \sum_{j=1}^{t_i(S)} \frac{1}{j}$$

The harmonic weighting promotes proportional representation and prevents dominance by heavily represented voters.

- **Fairness Modeling**

Representation shortfall penalty:

where $short(G, S) = \max(0, r_G - |S \cap G|)$

$$Pen_{rep}(S) = \sum_{G \in \mathcal{G}} short(G, S)$$

Worst-group satisfaction: $MM(S) = \min_{G \in \mathcal{G}} \bar{U}_G(S)$ which maximizes the satisfaction of the least represented group.

Our Approach

👉 OPTIMIZATION OBJECTIVE:

$$F(S) = U(S) - \lambda \cdot \text{Pen}_{rep}(S) + \mu \cdot MM(S)$$

- $U(S) \rightarrow$ approval-based utility,
- $\text{Pen}_{rep}(S) \rightarrow$ representation shortfall penalty,
- $MM(S) \rightarrow$ worst-group satisfaction.

👉 GREEDY COMMITTEE CONSTRUCTION:

PAF-Comm constructs the committee iteratively by selecting candidates that provide the best combined improvement in utility and fairness.

$$\text{score}(c) = \Delta U - \lambda \Delta \text{Pen}_{rep} + \mu \Delta MM$$

- $\Delta U \rightarrow$ increase in approval utility,
- $\Delta \text{Pen}_{rep} \rightarrow$ reduction in fairness shortfall,
- $\Delta MM \rightarrow$ improvement in worst-group satisfaction.

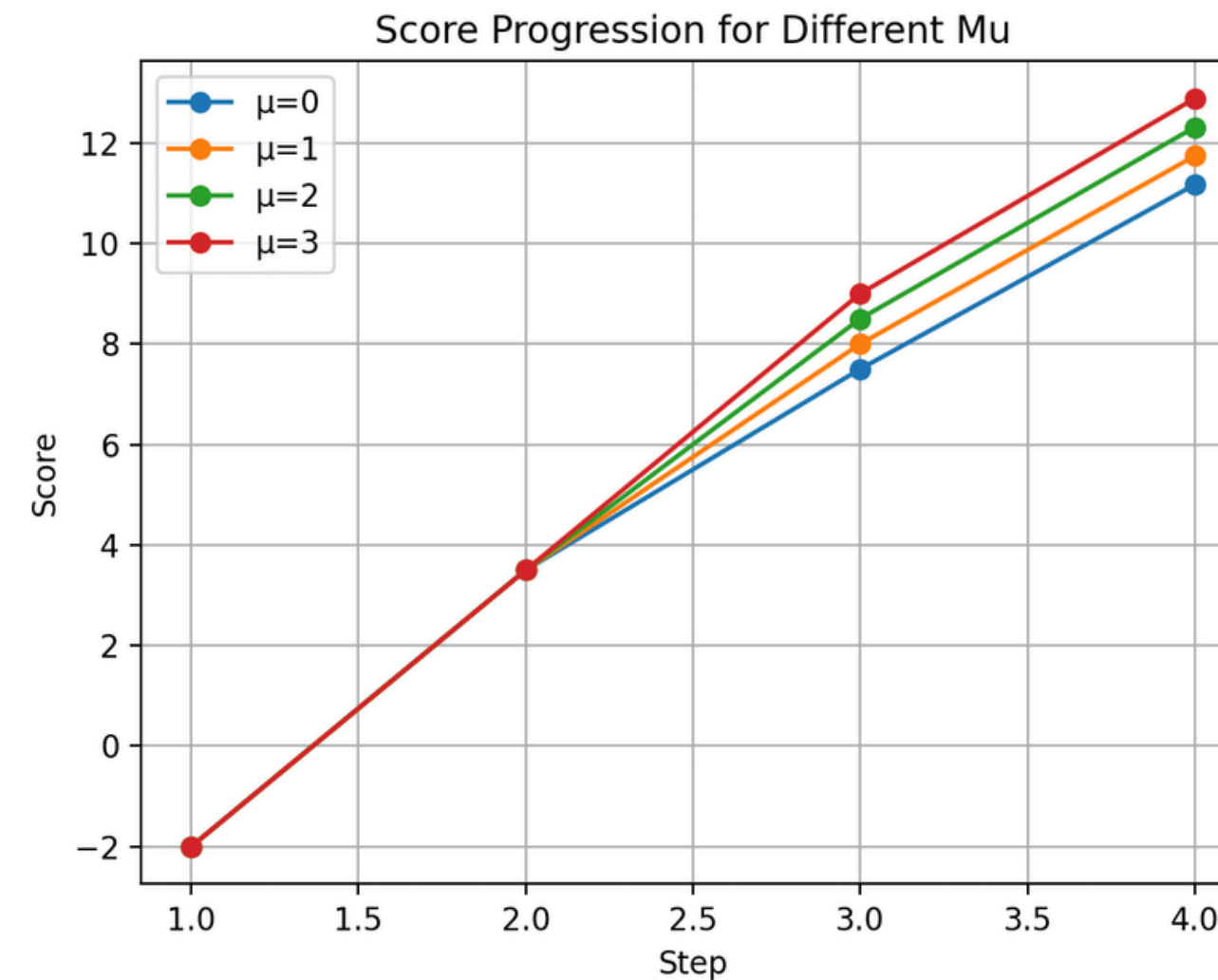
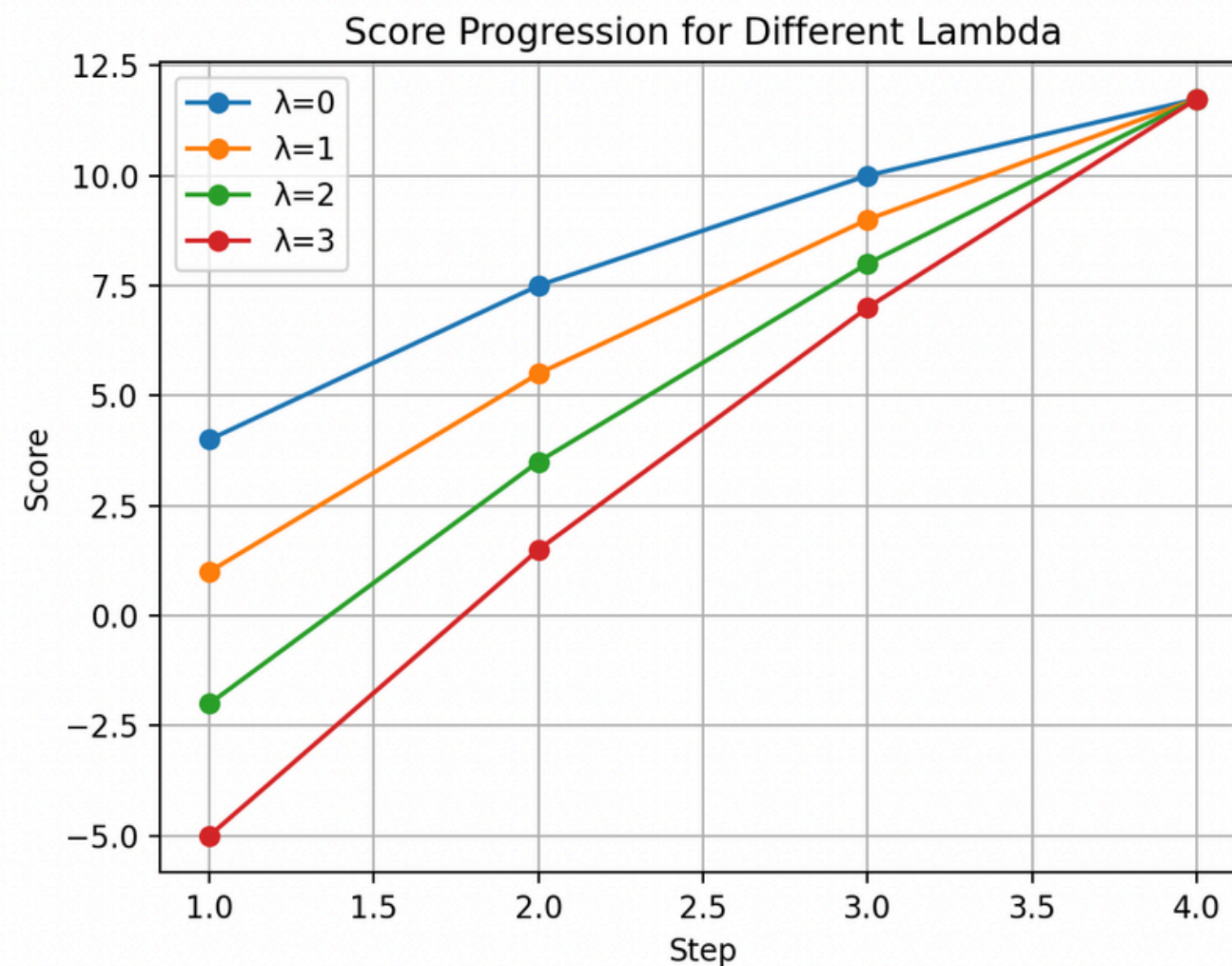
Results



SYNTHETIC DATASET:

- 6 candidates
- 7 voters.
- **Committee size:** $k=4$
- **Constraints:** $R_g = \{\text{Black Male: 1, White Female: 1, Black Female: 2}\}$

- Attribute Groups:
 - Male/Female
 - White/Black
- Intersectional groups:
 - Black Male, White Male
 - Black Female, White Female

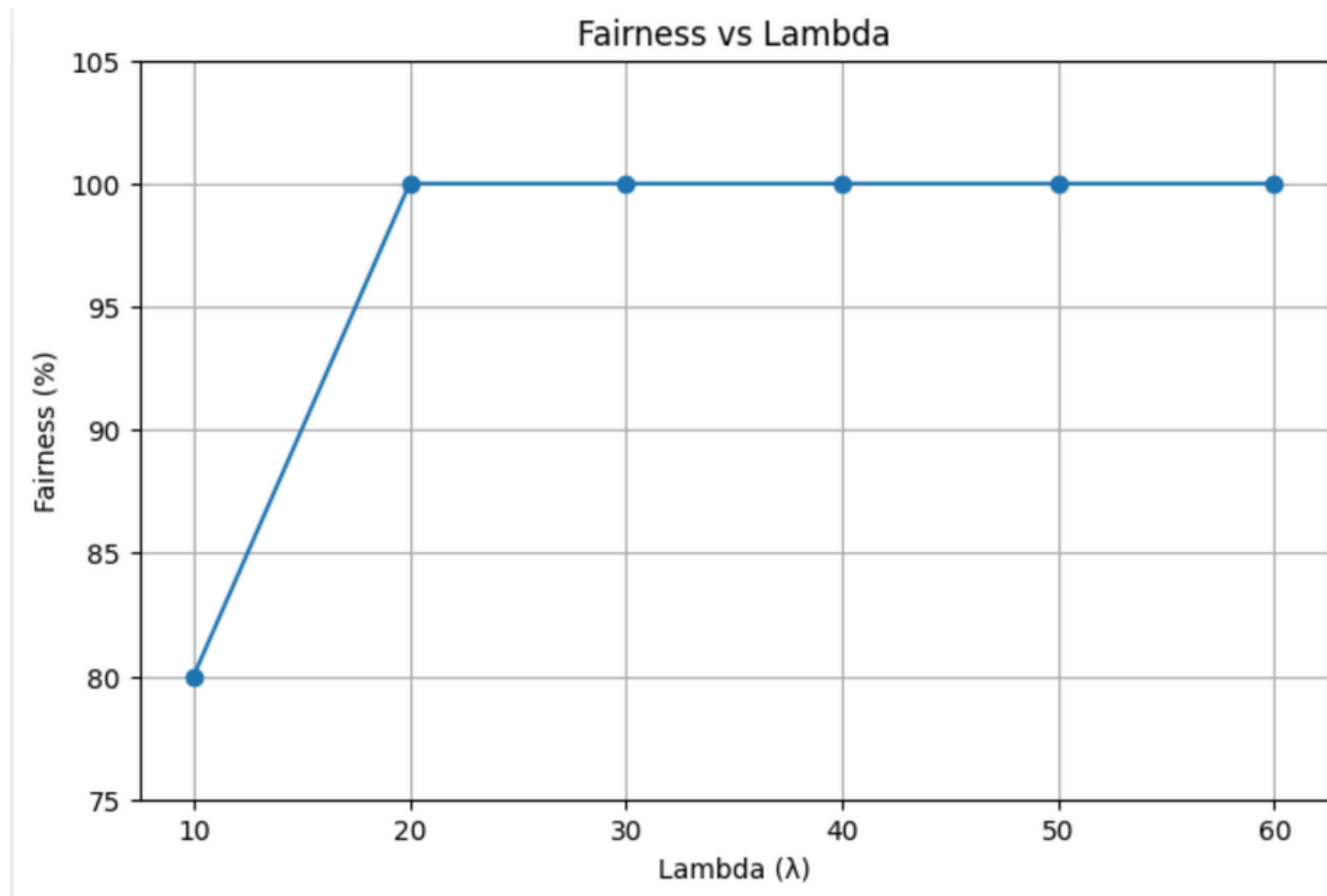


Results



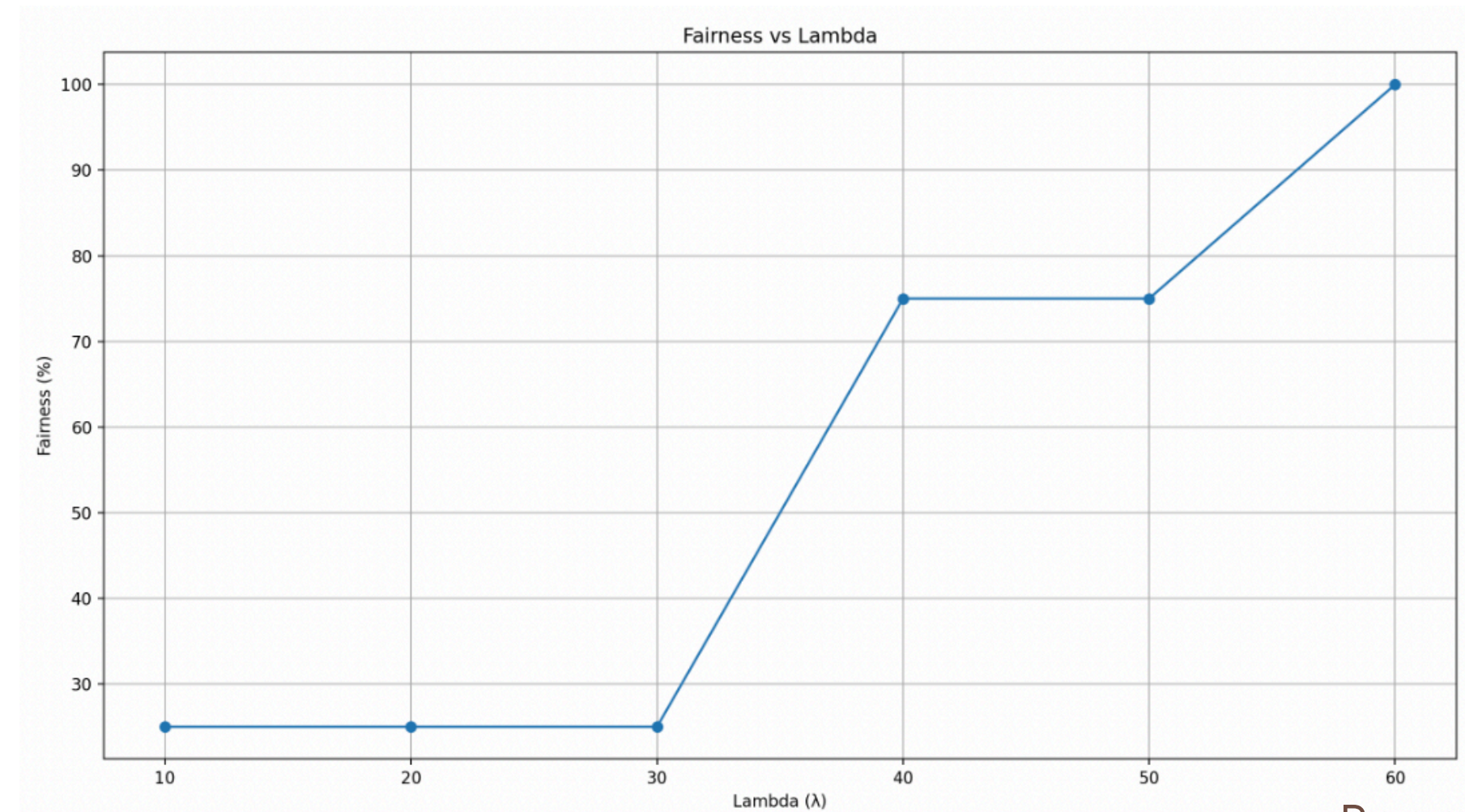
GOODREAD BOOKS DATASET:

- 10,000 candidates (books).
- 51,296 approval ballots.
- **Attributes:** Year Group, Popularity Group, Author Group
- **Multi-level and 3-way intersectional constraints.**
- **Committee size:** $k=8$



MOVIE LENS DATASET:

- 1682 candidates (movies).
- 943 approval ballots.
- **Attributes:** Genre, Year Group (Old/New)
- **Includes intersectional groups:** Horror_Old, Comedy_New
- **Committee size:** $k=5$



Local search refinement:

Local Search Direction:

- Estimate the price of fairness by performing local swaps/substitutions and measuring utility loss from the optimal committee.
- Limitations:
 - The original paper “Satisfying Complex Top-k Fairness Constraints by Preference Substitutions” considers settings where each voter provides a single preference.
 - Our setting involves multi-approval ballots.
 - Swapping candidates/ballots under multiple approvals and attribute constraints becomes computationally difficult.
 - Extending local substitution arguments to multi-level attribute constraints led to NP-hard formulations.

Marginal Quota Based Approach:

- To simplify the multi-level attribute setting, we first restricted the model to two binary attributes per candidate.
- Candidates were partitioned into four disjoint types:

$$C_{00}, C_{01}, C_{10}, C_{11}$$

01

COMMITTEE REPRESENTATION:

Committee is represented using a count vector:

$$x = (x_{00}, x_{01}, x_{10}, x_{11})$$

subject to

$$x_{00} + x_{01} + x_{10} + x_{11} = k$$

with quota constraints on each attribute group.

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APPROACH:

- Committee construction now becomes a **partitioning / count-allocation problem**.
- We enumerate feasible type-count vectors and select top-scoring candidates inside each partition.

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LIMITATION:

- This approach is tractable only for a fixed number of binary attributes
- For generalized ppp-attribute settings, the partitioning formulation becomes NP-hard via reduction from Set Cover

Constraint-Integrated Based

Constraint-Integrated Greedy Approval Voting (CIGAV) is a constrained approval-based committee selection framework that balances voter approval and attribute quota satisfaction.

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FRAMEWORK:

Candidate Selection Rule

At each iteration, candidate c_i is selected using: $Score(c_i) = \alpha \cdot AV(c_i) + \beta \cdot \sum_{t=1}^p s_i^t \cdot Urgency(\gamma_t)$
where:

- $AV(c_i)$: approval utility,
- s_i^t : contribution of candidate c_i to constraint γ_t
- $Urgency(\gamma_t)$: remaining quota pressure.

Limitation

- Greedy candidate selection may eliminate future feasible combinations during early stages.
- The algorithm may fail to construct a feasible committee under complex overlapping intersectional constraints, even when a feasible committee exists.
- Performance depends heavily on the ordering and urgency of constraints

Future Work

Feasibility-Aware Optimization:

- Develop stronger feasibility-preserving optimization techniques for intersectional constraints.

Algorithm:

- Design approximation or exact algorithms with theoretical guarantees for feasible committee construction.

References



Research Papers:

- MANI-Rank: Multiple Attribute and Intersectional Group Fairness for Consensus Ranking
- Satisfying Complex Top- k Fairness Constraints by Preference Substitutions
- Multiwinner Voting with Restricted Admissible Sets: Complexity and Strategyproofness
- Well-Structured Committees
- Multiwinner Voting with Fairness Constraints
- Locator: Local Stability for Rankings
- Fair Rank Aggregation
- Rank Aggregation with Proportionate Fairness

Q & A